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RESEARCH ARTICLE

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Linear versus cubic regression models for analyzing generalized reverse degree based topological indices of certain latest corona treatment drug molecules

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[Correction added after first online publication on 5 July 2023. Author biography has been removed from the article.]

Abstract

The major challenges encountered by medical researchers in developing new drugs are time consumption, increased cost, establishing a safety profile for the drugs, poor solubility, and inadequate experimental data. In its theoretical aspects, chemical graph theory plays a vital role in drug design and development by analyzing the structural parameters of molecules. Topological indices aim at the mathematical representation of a molecular structure, which is used to analyze the effectiveness of drugs and enhance the drug development process. In this study, we consider certain recently used drugs such as dexamethasone, molnupiravir, nirmatrelvir, ivermectin, ribavirin, baricitinib, favipiravir, duvelisib, L-ascorbic acid, sofosbuvir, remdesivir, and pioglitazone for omicron, delta and other variants of coronaviruses. For these drug molecules, we propose a generalized form of reverse degree parameters and compute their associated topological indices with limiting behaviors. We undertake QSPR study on the potential of generalized reverse-degree indices using linear and cubic regression models.

KEYWORDS
COVID-19 drugs, quantitative structure property relationship, reverse degree edge partition: topological indices

1 | INTRODUCTION

The entire world aims to discover the exact medications for COVID-19, while medical researchers have been attempting to eradicate it with existing anti-malarial, anti-viral, and anti-inflammatory drugs that have been widely used for many years and are easily available at low cost, such as hydroxychloroquine, chloroquine, remdesivir, and corticosteroids [1]. However, not all of the outcomes of such drug treatment were analytically significant. Some of them helped to accelerate the recovery time of patients with COVID-19 disease, and some of them did not have any positive outcome due to the unique immune systems of human beings that may react differently to different drugs. Consequently, there are no specific treatments for this disease, but early diagnosis can be treated with antiviral drugs and oxygen supplies. L-ascorbic acid, an antioxidant, plays a vital role in boosting immunity and enhancing oxygen levels in the body [2]. However, they cannot completely eradicate the disease since the coronavirus keeps on transforming into new variants such as alpha, beta, gamma, delta, and the Omicron variant, which makes it challenging to discover a cure. The first version of omicron BA_1 had dozens of changes in its genetic makeup and BA_2 quickly followed with enough genetic modification to get around our immune defense system, followed by the most distinct versions BA_4 and BA_5 [3,4]. Compared to other coronaviruses, the Omicron variant is more transmissible, resistant, and vulnerable.

Dexamethasone, a corticosteroid with a low to moderate dosage, has a lower mortality rate if given at that stage of disease where there is inflammation and is not recommended for mild symptoms [5]. The study demonstrated that molnupiravir and nirmatrelvir effectively inhibited

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Omicron variant infection. The use of molnupiravir and nirmatrelvir plays a vital role in treating Omicron-infected patients [6]. Nirmatrelvir is also used to treat symptomatic severe COVID-19, which lowers the risk by 89% [7,8]. In addition to dexamethasone and remdesivir, baricitinib has considerably enhanced health and decreased mortality risk in hospitalized adults patients with COVID-19 disease [9]. Ribavirin is widely available at a minimum cost and has a significant impact on COVID-19 [10]. Favipiravir plays an important role in fighting coronavirus variants [11] and sofosbuvir is used to treat outpatients with mild infection [12]. Using ivermectin at the start of an infection may help prevent it from progressing to a severe disease and has clearly shown a significant reduction in COVID-19 deaths. Because of its apparent safety and low cost, ivermectin is expected to have a considerable global impact on COVID-19 disease [13].

Topological indices, also known as molecular descriptors, are mathematically related formulas under molecular characteristics that can be used in the analysis of physical and chemical properties and are then used by medical researchers for drug development as it is a low-cost and quick-processing computational tool. Topological indices play a significant role in mathematical chemistry, more specifically in QSAR and QSPR analysis. There are numerous types of topological indices at present, including distance-based, degree-based, and eccentricity-based topological indices, which are widely used for chemical structure analysis. Wiener proposed the first and most popular topological index in 1947 [14] known as the Wiener index which is used to calculate the boiling point of chemical structures. The first degree-based topological index was proposed by Gutman in 1972 [15] known as Zagreb index but it was used for a completely different process that was latter included [16]. After that, numerous indices were presented and thoroughly examined, including Randić, atom bond connectivity, geometric-arithmetic, and so forth. Motivated by such degree-type indices, Kulli introduced the reverse degree and investigated topological indices derived from it [17]. Jung et al. [18] proposed reverse atom bond connectivity index and reverse geometric arithmetic index, and then reverse degree indices have attracted many researchers for computing chemical structures [19–21] and molecular drugs [22–24]. In recent papers [25–31] various distance/degree based topological indices of dexamethasone, molnupiravir, baricitinib, ribavirin, pioglitazone, favipiravir, nirmatrelivir, and temperature based topological indices [32] of sofosbuvir, ribavirin were computed and the reverse degree-based topological indices of certain COVID-19 drugs were discussed [33].

In general the chemical structure of drugs are converted into 2D/3D graph structure. Throughout this paper, the graph G denotes the hydrogen depleted graph obtained from the molecular structure where atoms are marked to the vertices and bonds to the edges. Let $V(G)$ and $E(G)$ represent the vertex and edge sets of the molecular drug G respectively. The number of edges in the drug G that incident to the vertex v is called degree and it is denoted by d_v. The maximum degree of G is denoted by $\Delta(G)$. By combining the degree and the maximum degree of G, Kulli [17] proposed the reverse degree and defined as $\mathfrak{R}d_v = \Delta(G) - d_v + 1$. It is natural to generalize the reverse degree by adding a new variable k in the following and we call it as modified reverse degree under k.

$$M_k\mathfrak{R}(d_v) = \begin{cases} \Delta(G) - d_v + k & : k \le d_v \\ \Delta(G) - d_v + k(mod\,\Delta(G)) & : k > d_v \end{cases}$$

It is important to note that the degree and reverse degree of the vertices are fixed measures whereas the modified reverse degree of the vertices are variable measures with respect to reverse degree parameter k. Clearly, the investigation of modified reverse degree indices is more important than that of degree indices. Therefore, in this paper, we propose the modified reverse atom bond connectivity index $[M_k\mathfrak{R}ABC(G)]$, modified reverse geometric-arithmetic index $[M_k\mathfrak{R}GA(G)]$, modified reverse arithmetic-geometric index $[M_k\mathfrak{R}AG(G)]$, modified reverse Zagreb-1 index $[M_k\mathfrak{R}M_1(G)]$, modified reverse Zagreb-2 index $[M_k\mathfrak{R}M_2(G)]$, modified reverse Randić index $[M_k\mathfrak{R}R_{\frac{1}{2}}(G)]$, modified reverse forgotten index $[M_k\mathfrak{R}F(G)]$, modified reverse redefined Zagreb-1 $[M_k\mathfrak{R}ReZ_1(G)]$, redefined Zagreb-2 $[M_k\mathfrak{R}ReZ_2(G)]$and redefined Zagreb-3 $[M_k\mathfrak{R}ReZ_3(G)]$.

$$M_k\mathfrak{R}ABC(G) = \sum_{uv \in E(G)} \sqrt{\frac{M_k\mathfrak{R}(d_u) + M_k\mathfrak{R}(d_v) - 2}{M_k\mathfrak{R}(d_u) \times M_k\mathfrak{R}(d_v)}} \tag{1}$$

$$M_k\mathfrak{R}GA(G) = \sum_{uv \in E(G)} \frac{2\sqrt{M_k\mathfrak{R}(d_u) \times M_k\mathfrak{R}(d_v)}}{M_k\mathfrak{R}(d_u) + M_k\mathfrak{R}(d_v)} \tag{2}$$

$$M_k\mathfrak{R}AG(G) = \sum_{uv \in E(G)} \frac{M_k\mathfrak{R}(d_u) + M_k\mathfrak{R}(d_v)}{2\sqrt{M_k\mathfrak{R}(d_u) \times M_k\mathfrak{R}(d_v)}} \tag{3}$$

$$M_k\mathfrak{R}M_1(G) = \sum_{uv \in E(G)} [M_k\mathfrak{R}(d_u) + M_k\mathfrak{R}(d_v)] \tag{4}$$

$$M_k\mathfrak{R}M_2(G) = \sum_{uv \in E(G)} [M_k\mathfrak{R}(d_u) \times M_k\mathfrak{R}(d_v)] \tag{5}$$

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$$M_k\mathfrak{R}R_{\frac{1}{2}}(G)=\sum_{uv\in E(G)}[M_k\mathfrak{R}(d_u)\times M_k\mathfrak{R}(d_v)]^{\frac{1}{2}} \tag{6}$$

$$M_k\mathfrak{R}F(G)=\sum_{uv\in E(G)}[(M_k\mathfrak{R}(d_u))^2+(M_k\mathfrak{R}(d_v))^2] \tag{7}$$

$$M_k\mathfrak{R}ReZ_1(G)=\sum_{uv\in E(G)}\frac{M_k\mathfrak{R}(d_u)+M_k\mathfrak{R}(d_v)}{M_k\mathfrak{R}(d_u)\times M_k\mathfrak{R}(d_v)} \tag{8}$$

$$M_k\mathfrak{R}ReZ_2(G)=\sum_{uv\in E(G)}\frac{M_k\mathfrak{R}(d_u)\times M_k\mathfrak{R}(d_v)}{M_k\mathfrak{R}(d_u)+M_k\mathfrak{R}(d_v)} \tag{9}$$

$$M_k\mathfrak{R}ReZ_3(G)=\sum_{uv\in E(G)}[M_k\mathfrak{R}(d_u)+M_k\mathfrak{R}(d_v)][M_k\mathfrak{R}(d_u)\times M_k\mathfrak{R}(d_v)] \tag{10}$$

The paper is organized as follows: Section 2 discusses the computation of modified reverse degree-based topological indices of various recent COVID-19 drug molecules under limiting behaviors of the reverse degree. In Section 3, we analyze the QSPR models for the modified reverse degree indices with their physicochemical properties for COVID-19 drug molecules. We conclude the paper in Section 4.

2 | METHODOLOGY

To compute the modified reverse degree-based topological descriptors of the recently used COVID-19 drug molecules (see Figure 1), we first present edge partition by enumerating the original degree of the drug molecules' bond end vertices. Then we transform the original degree into a modified reverse degree as defined in the previous section, and following that, we carry over the computation process.

1. For the molecular structure of drugs such as ribavirin, favipiravir, molnupiravir, L-ascorbic acid, duvelisib, pioglitazone which have a maximum degree of 3, the modified reverse degree is as follows:

$$M_1\mathfrak{R}(d_v)=\begin{cases}3 & :d_v=1\\ 2 & :d_v=2\\ 1 & :d_v=3\end{cases}$$

$$M_2\mathfrak{R}(d_v)=\begin{cases}1 & :d_v=1\\ 3 & :d_v=2\\ 2 & :d_v=3\end{cases}$$

$$M_3\mathfrak{R}(d_v)=\begin{cases}2 & :d_v=1\\ 1 & :d_v=2\\ 3 & :d_v=3\end{cases}$$

2. For the molecular structure of drugs such as dexamethasone, nirmatrelvir, baricitinib, sofosbuvir, ivermectin, and remdesivir which have a maximum degree of four, the modified reverse degree can be found as above. For instance,

$$M_2\mathfrak{R}(d_v)=\begin{cases}1 & :d_v=1\\ 4 & :d_v=2\\ 3 & :d_v=3\\ 2 & :d_v=4\end{cases}$$

We now compute and present the modified reverse degree-based indices of drugs shown in Figure 1 using the degree edge partition from Table 1 along with reverse degree measures for $k=1,2,3$ and presented in Tables 2–4.

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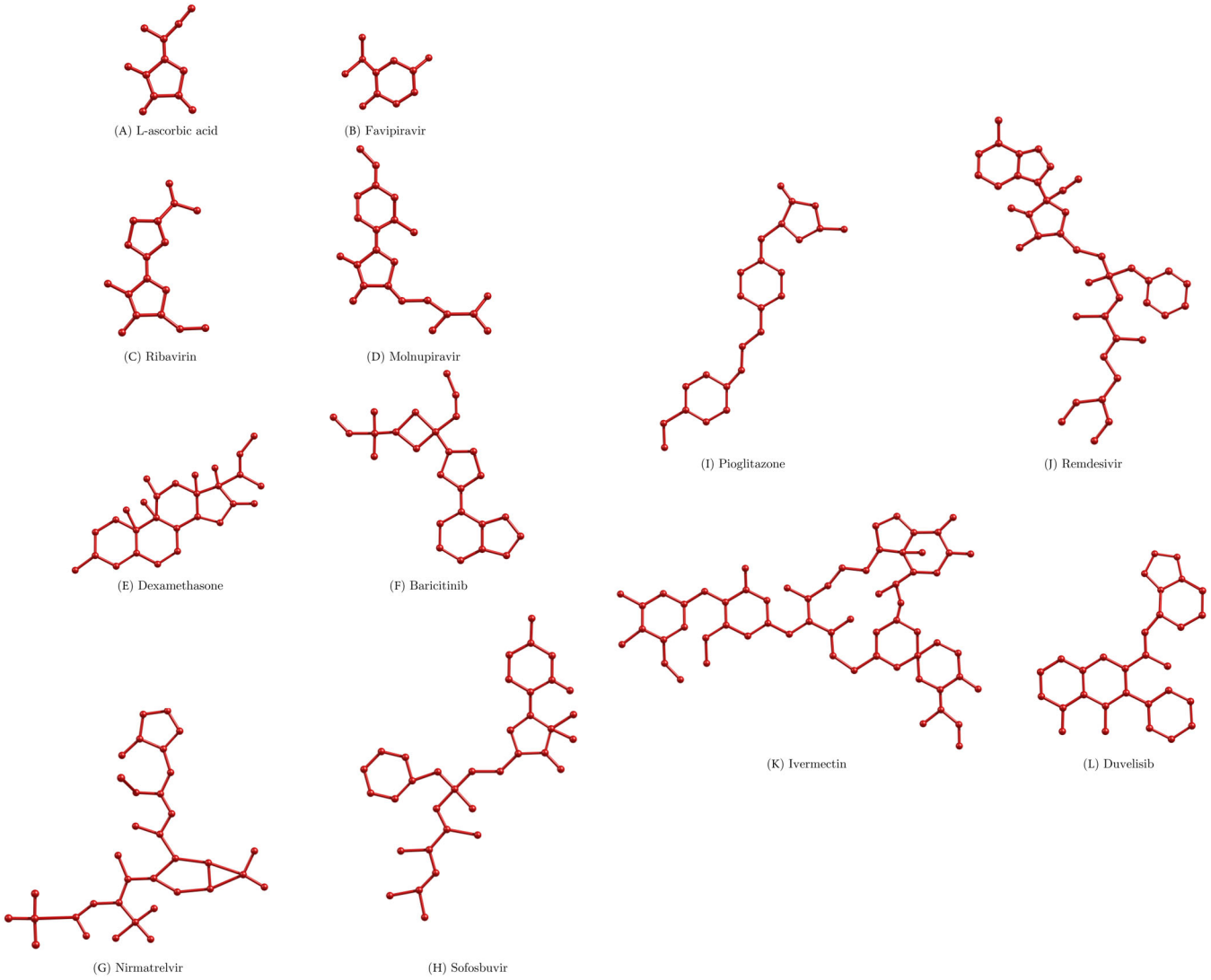


FIGURE 1 Molecular structure of COVID-19 drugs.

TABLE 1 Degree bond partitions for COVID-19 drugs.

		Degree-based bond partitions and frequencies								
Graphs	**Drugs**	**(1,2)**	**(1,3)**	**(1,4)**	**(2,2)**	**(2,3)**	**(2,4)**	**(3,3)**	**(3,4)**	**(4,4)**
G_1	Dexamethasone	1	4	4	2	9	2	1	6	2
G_2	Remdesivir	3	5	1	9	14	5	6	2	0
G_3	Ivermectin	3	10	1	5	31	4	11	3	0
G_4	Nirmatrelvir	1	4	8	2	11	0	7	4	0
G_5	Sofosbuvir	0	7	3	6	13	3	4	2	0
G_6	Baricitinib	2	0	2	6	10	4	3	2	0
G_7	Molnupiravir	1	6	0	2	9	0	6	0	0
G_8	Ribavirin	1	4	0	1	7	0	5	0	0
G_9	Pioglitazone	1	2	0	6	17	0	1	0	0
G_{10}	L-ascorbic acid	1	4	0	0	3	0	4	0	0
G_{11}	Duvelisib	0	3	0	10	12	0	9	0	0
G_{12}	Favipiravir	0	4	0	1	4	0	2	0	0

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TABLE 2 Modified reverse degree topological indices of COVID-19 drugs with reverse parameter $k = 1$.

Drugs	$M_1\mathfrak{R}GA$	$M_1\mathfrak{R}AG$	$M_1\mathfrak{R}ABC$	$M_1\mathfrak{R}M_2$	$M_1\mathfrak{R}R_{\frac{1}{2}}$	$M_1\mathfrak{R}F$	$M_1\mathfrak{R}M_1$	$M_1\mathfrak{R}ReZ_1$	$M_1\mathfrak{R}ReZ_2$	$M_1\mathfrak{R}ReZ_3$
Dexamethasone	32.939	33.112	21.218	156	66.773	388	142	34.0833	31.5476	814
Molnupiravir	22.661	25.4948	13.38	56	35.567	146	76	36.667	16.7	182
Nirmatrelvir	28.01635	39.7225	27.5915	196	83.3791	496	178	37.0833	40.758	1010
Baricitinib	28.882	30.46705	20.8956	174	69.1797	396	144	27.333	33.36190	940
Ribavirin	17.043	16.5604	9.630	41	26.2772	106	56	27.667	12.3667	146
Favipiravir	10.235	10.86144	6.802	26	16.585	72	36	16.3333	7.6667	92
Duvelisib	32.912	35.1921	18.006	82	51.167	188	106	50	24.75	286
L-ascorbic acid	11.272	16.0034	6.094	28	17.620	76	38	18.667	8.2	104
Sofosbuvir	32.06	49.9830	27.432	229	91.667	540	192	34.8333	46.9166	1222
Pioglitazone	25.7340	28.36124	18.125	71	42.96	168	90	37	20.5333	254
Ivermectin	65.8635	70.32772	48.2927	413	164.782	928	340	60.5	79.9762	2194
Remdesivir	44.6038	46.768	34.8576	288	111.316	644	230	40.0833	53.993	1586

TABLE 3 Modified reverse degree topological indices of COVID-19 drugs with reverse parameter $k = 2$.

Drugs	$M_2\mathfrak{R}GA$	$M_2\mathfrak{R}AG$	$M_2\mathfrak{R}ABC$	$M_2\mathfrak{R}M_2$	$M_2\mathfrak{R}R_{\frac{1}{2}}$	$M_2\mathfrak{R}F$	$M_2\mathfrak{R}M_1$	$M_2\mathfrak{R}ReZ_1$	$M_2\mathfrak{R}ReZ_2$	$M_2\mathfrak{R}ReZ_3$
Dexamethasone	29.707	32.45	21.7324	233	81.116	518	180	21	30.062	1254
Molnupiravir	23.341	24.704	16.99	111	50.263	241	103	25.1667	24.55	522
Nirmatrelvir	35.6129	38.5506	25.60917	283	97.145	614	200	37.33	47.2905	1794
Baricitinib	28.1139	30.008	19.5181	299	90.6821	646	186	21	42.8095	2074
Ribavirin	17.496	18.542	12.797	82	37.5353	179	77	18.833	18.3167	380
Favipiravir	10.690	11.325	7.737	49	22.4548	106	46	12	10.967	230
Duvelisib	33.586	34.430	23.637	204	81.6365	423	165	30.167	40.4	1062
L-ascorbic acid	11.576	12.46	8.595	45	22.737	101	47	13.833	11.0167	190
Sofosbuvir	36.55	39.623	26.1047	351	110.785	760	228	31	53.936	2382
Pioglitazone	26.408	27.626	18.958	167	66.202	357	135	23.5	32.4833	874
Ivermectin	65.3957	70.98342	46.8255	645	203.784	1408	420	52	99.1429	4306
Remdesivir	43.2028	47.074	32.5414	447	137.613	978	284	34	68.3119	3078

The computational procedure for deriving the numerical values in Tables 2,3, and 4 is briefly explained in the following. For G_i $(1 \le i \le 12)$, $TI \in \{GA, AG, ABC, M_2, R_{\frac{1}{2}}, F, M_1, ReZ_1, ReZ_2, ReZ_3\}$ and $k = 1,2,3$, the general form of modified reverse degree topological index is defined by

$$M_k\mathfrak{R}TI(G_i) = \sum_{uv \in E(G_i)} TI(M_k\mathfrak{R}(d_u), M_k\mathfrak{R}(d_v))$$

where $TI(M_k\mathfrak{R}(d_u), M_k\mathfrak{R}(d_v))$ is the topological index function such that $TI(M_k\mathfrak{R}(d_u), M_k\mathfrak{R}(d_v)) = TI(M_k\mathfrak{R}(d_v), M_k\mathfrak{R}(d_u))$.

The drug compound dexamethasone G_1 has the maximum degree of four and applying the modified reverse degree in Table 1, we have

$$\begin{aligned}
M_1\mathfrak{R}TI(G_1) &= TI(4,3) + 4\,TI(4,2) + 4\,TI(4,1) + 2\,TI(3,3) + 9\,TI(3,2) + 2\,TI(3,1) \\
&\quad + TI(2,2) + 6\,TI(2,1) + 2\,TI(1,1), \\
M_2\mathfrak{R}TI(G_1) &= TI(1,4) + 4\,TI(1,3) + 4\,TI(1,2) + 2\,TI(4,4) + 9\,TI(4,3) + 2\,TI(4,2) \\
&\quad + TI(3,3) + 6\,TI(3,2) + 2\,TI(2,2), \\
M_3\mathfrak{R}TI(G_1) &= TI(2,1) + 4\,TI(2,4) + 4\,TI(2,3) + 2\,TI(1,1) + 9\,TI(1,4) + 2\,TI(1,3) \\
&\quad + TI(4,4) + 6\,TI(4,3) + 2\,TI(3,3).
\end{aligned}$$

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TABLE 4 Modified reverse degree topological indices of COVID-19 drugs with reverse parameter $k = 3$.

Drugs	$M_3\Re GA$	$M_3\Re AG$	$M_3\Re ABC$	$M_3\Re M_2$	$M_3\Re R_{\frac{1}{2}}$	$M_3\Re F$	$M_3\Re M_1$	$M_3\Re ReZ_1$	$M_3\Re ReZ_2$	$M_3\Re ReZ_3$
Dexamethasone	28.5037	34.007	21.610	208	76.775	424	166	26.25	22.5	654
Molnupiravir	22.616	25.5767	16.30	121	51.70	285	109	26.5	24.616	622
Nirmatrelvir	34.31139	40.2597	25.5873	288	98.1803	704	210	34.75	46.25714	1918
Baricitinib	26.289	24.359	17.882	146	59.5838	404	132	37.1667	27.16190	884
Ribavirin	16.924	19.226	12.584	93	39.337	219	83	19.5	18.716	482
Favipiravir	10.383	11.7013	7.4278	55	23.726	130	50	12	11.3	278
Duvelisib	32.332	35.918	17.919	145	65.133	341	137	44.5	31.1	740
L-ascorbic acid	11.46	12.607	8.6517	71	28.408	159	59	11.5	13.7167	378
Sofosbuvir	34.516	42.2213	24.52	229	87.272	464	192	43.1667	44.011	1334
Pioglitazone	24.625	29.732	16.668	80	44.758	231	99	38.5	20.3167	336
Ivermectin	60.4696	77.2091	48.685	445	163.297	756	364	74.1667	73.9762	2866
Remdesivir	41.0319	49.8001	27.5360	252	97.4227	676	214	55.41667	44.745	1636

TABLE 5 Physicochemical properties of drug molecules.

Drugs	Polarization	Boiling point	Enthalpy of vaporization	Molar volume	Heavy atom count	Molecular refraction	Molar weight	Complexity
Dexamethasone	39.7	568.2	98	296.2	28	100.2	392.5	805
Molnupiravir	29.4	—	—	202.8	23	74.1	329.31	534
Nirmatrelvir	46.3	742.4	108.3	394.1	35	116.9	499.5	964
Baricitinib	38.9	707.2	103.5	238.1	26	98.2	371.4	678
Ribavirin	20.3	639.8	99.3	117.1	17	51.2	244.20	304
Favipiravir	13.2	552.6	86.4	97.2	11	33.2	157.10	282
Duvelisib	46	—	—	282.8	30	116	416.90	668
L-ascorbic acid	14	552.7	95.8	90.1	12	35.3	176.12	232
Sofosbuvir	48.9	—	—	374.6	36	123.5	529.5	913
Pioglitazone	38.9	575.4	86.2	282.8	25	98.2	356.40	466
Ivermectin	91.5	940.4	155.2	708	62	230	875.1	1680
Remdesivir	59.3	—	—	409	42	149.5	602.6	1010

When $TI = GA$, we have $GA(4,3) = \frac{2\sqrt{12}}{7}$, $GA(4,2) = \frac{2\sqrt{8}}{6}$, $GA(4,1) = \frac{2\sqrt{4}}{5}$, $GA(3,3) = \frac{2\sqrt{9}}{6}$, $GA(3,2) = \frac{2\sqrt{6}}{5}$, $GA(3,1) = \frac{2\sqrt{3}}{4}$, $GA(2,2) = \frac{2\sqrt{4}}{4}$, $GA(2,1) = \frac{2\sqrt{2}}{3}$, $GA(1,1) = 1$, and so forth, we obtain $M_1\Re GA(G_1) = 32.939$, $M_2\Re GA(G_1) = 29.707$, and $M_3\Re GA(G_1) = 28.5037$. In a similar way, the other indices as well as for other drug compounds with maximum degree four can be computed.

The drug compounds with maximum degree three, for instance, the molnupiravir G_7, we have

$$\begin{aligned} M_1\Re TI(G_7) &= TI(3,2) + 6\,TI(3,1) + 2\,TI(2,2) + 9\,TI(2,1) + 6\,TI(1,1), \\ M_2\Re TI(G_7) &= TI(1,3) + 6\,TI(1,2) + 2\,TI(3,3) + 9\,TI(3,2) + 6\,TI(2,2), \\ M_3\Re TI(G_7) &= TI(2,1) + 6\,TI(2,3) + 2\,TI(1,1) + 9\,TI(1,3) + 6\,TI(3,3). \end{aligned}$$

Suppose $TI = F$, we have $F(3,2) = 13$, $F(3,1) = 10$, $F(2,2) = 8$, $F(2,1) = 5$, $F(1,1) = 2$, and so forth, we get $M_1\Re F(G_7) = 146$, $M_2\Re F(G_7) = 241$, and $M_3\Re F(G_7) = 285$. Other indices and drug compounds with a maximum degree of three can be computed in the same way.

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TABLE 6 Correlation between modified reverse degree based indices (reverse parameter $k = 1$) with physical and chemical properties of drugs.

Properties	Polarization	Molar volume	Heavy atom count	Boiling point	Enthalpy of vaporization	Molecular refraction	Molar weight	Complexity
$M_1\mathfrak{R}ABC$	0.978	**0.981**	**0.987**	0.904	0.893	0.978	**0.989**	0.980
$M_1\mathfrak{R}GA$	**0.985**	0.956	0.975	0.862	0.902	**0.985**	0.970	0.954
$M_1\mathfrak{R}AG$	0.975	0.975	0.984	0.894	0.909	0.976	0.987	0.970
$M_1\mathfrak{R}M_1$	0.964	0.968	0.979	0.919	0.924	0.964	0.981	**0.984**
$M_1\mathfrak{R}M_2$	0.934	0.939	0.952	0.923	**0.931**	0.934	0.956	0.965
$M_1\mathfrak{R}R_{\frac{1}{2}}$	0.966	0.968	0.979	0.921	0.928	0.966	0.982	0.983
$M_1\mathfrak{R}F$	0.926	0.939	0.948	0.915	0.920	0.926	0.953	0.968
$M_1\mathfrak{R}ReZ_1$	0.891	0.855	0.871	0.830	0.841	0.891	0.860	0.811
$M_1\mathfrak{R}ReZ_2$	0.967	0.968	0.980	**0.925**	0.928	0.967	0.982	0.983
$M_1\mathfrak{R}ReZ_3$	0.910	0.915	0.930	0.920	0.930	0.909	0.879	0.949

TABLE 7 Correlation between modified reverse degree based indices (reverse parameter $k = 2$) with physical and chemical properties of drugs.

Properties	Polarization	Molar volume	Heavy atom count	Boiling point	Enthalpy of vaporization	Molecular refraction	Molar weight	Complexity
$M_2\mathfrak{R}ABC$	0.995	0.984	0.998	0.901	0.911	0.995	0.996	0.977
$M_2\mathfrak{R}GA$	**0.997**	**0.986**	**0.998**	0.906	0.911	**0.997**	**0.996**	0.978
$M_2\mathfrak{R}AG$	0.993	0.982	0.997	0.898	0.895	0.993	0.996	0.975
$M_2\mathfrak{R}M_1$	0.986	0.974	0.989	0.917	0.919	0.986	0.989	**0.978**
$M_2\mathfrak{R}M_2$	0.966	0.953	0.971	0.922	0.919	0.967	0.973	0.966
$M_2\mathfrak{R}R_{\frac{1}{2}}$	0.987	0.974	0.989	0.917	0.918	0.987	0.989	0.977
$M_2\mathfrak{R}F$	0.964	0.952	0.969	0.920	0.921	0.964	0.972	0.966
$M_2\mathfrak{R}ReZ_1$	0.945	0.957	0.963	0.924	0.893	0.945	0.961	0.936
$M_2\mathfrak{R}ReZ_2$	0.983	0.966	0.985	**0.944**	**0.928**	0.983	0.986	0.961
$M_2\mathfrak{R}ReZ_3$	0.940	0.891	0.927	0.935	0.918	0.940	0.883	0.903

3 | QSPR ANALYSIS

A quantitative structure-activity relationship (QSAR) is a mathematical model that examines the connection between a molecular compound's structural and biological or physicochemical properties. The quantitative structure-property relationship (QSPR) uses a compound's molecular structure to make predictions about its physical and chemical properties [34–40]. Topological indices play a considerable role in QSPR analysis, which is used to correlate the various physicochemical properties of drugs used for the treatment of COVID-19, including Omicron variants. Since the 1960s, computer technology has demonstrated great promise in assisting chemists in the development of new drugs. Computer-based analysis of drugs is predominant for fabricating, manufacturing, and re-designing molecular compounds. In computer-aided drug design, mathematical tools are used to monitor the biological activity, metabolism, and relationship between the drug and a molecular structure. Structure-based drug design and ligand-based drug design are the two main types of computer-aided drug design. QSPR analysis is a ligand-based drug design method that was proposed by Hanch and Fujitha in 1964 [41]. Out of many molecular descriptors, topological indices are extensively used in QSPR analysis pertaining to drug investigations now [42–48].

We now study the effectiveness of modified reverse degree indices by QSPR analysis for certain physical and chemical properties such as polarization, boiling point, enthalpy of vaporization, heavy atom count, molar volume, molar refraction, molar weight, and complexity of COVID-19 drug molecules (see Figure 1) which were collected from PubChem (https://pubchem.ncbi.nlm.nih.gov) and ChemSpider (http://www.chemspider.com) as shown in Table 5.

We have presented the correlation coefficient between the modified reverse degree indices and the physicochemical properties of COVID-19 drugs by varying the reverse degree parameters in Tables 6,7, and 8. It is clear from these tables that when the reverse degree parameters

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TABLE 8 Correlation between modified reverse degree based indices (reverse parameter $k = 3$) with physical and chemical properties of drugs.

Properties	Polarization	Molar volume	Heavy atom count	Boiling point	Enthalpy of vaporization	Molecular refraction	Molar weight	Complexity
$M_3\Re ABC$	0.973	0.986	0.985	0.908	0.933	0.973	0.984	0.986
$M_3\Re GA$	**0.995**	0.984	**0.997**	0.905	0.909	**0.995**	**0.995**	0.978
$M_3\Re AG$	0.987	**0.988**	0.994	0.881	0.909	0.987	0.993	0.973
$M_3\Re M_1$	0.964	0.981	0.983	0.910	0.931	0.964	0.982	**0.995**
$M_3\Re M_2$	0.920	0.952	0.951	0.896	0.918	0.920	0.950	0.982
$M_3\Re R_{\frac{1}{2}}$	0.961	0.978	0.981	0.907	0.927	0.961	0.980	0.994
$M_3\Re F$	0.871	0.891	0.904	0.860	0.806	0.872	0.907	0.924
$M_3\Re ReZ_1$	0.973	0.930	0.955	0.886	0.855	0.973	0.953	0.896
$M_3\Re ReZ_2$	0.946	0.961	0.971	**0.961**	**0.943**	0.946	0.974	0.965
$M_3\Re ReZ_3$	0.895	0.926	0.930	0.953	0.932	0.894	0.853	0.947

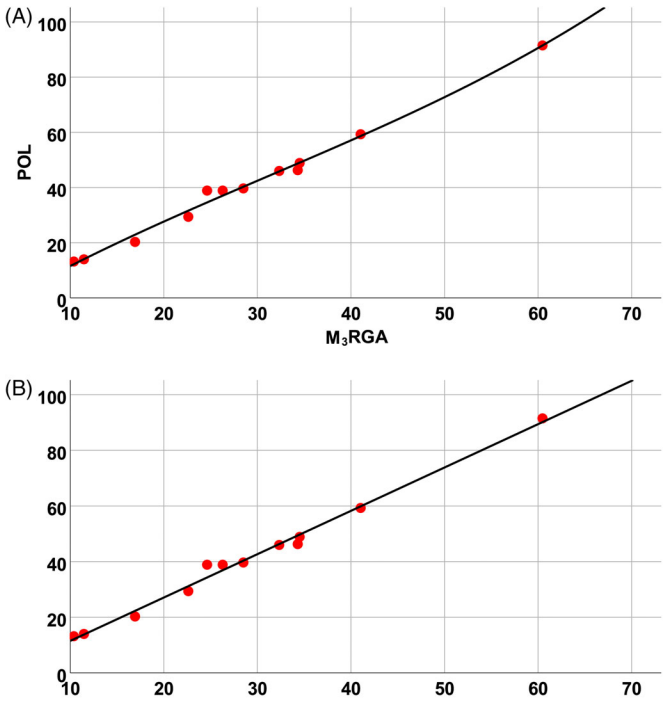


FIGURE 2 Cubic regression graph versus linear regression graph for polarization. (A) Cubic model. (B) Linear model.

increase, the corresponding correlation is high. Hence, in the following section, we develop the regression models for only the high correlation reverse degree parameter.

3.1 | Curvilinear regression analysis

In this section, we determine the best possible linear and cubic regression models for predicting physicochemical properties of COVID-19 drugs. Let $P = a\,(TI) + d$ be the linear regression equation and $P = a\,(TI)^3 + b\,(TI)^2 + c\,(TI) + d$ be the cubic regression equation where P represents physical property, a, b, and c represent regression coefficients and d represents regression constant. The analysis is based on r^2 value and standard error (SE).

The linear versus cubic regression models derived based on the reverse degree parameter $k = 3$ are given in Table 9. The comparison graphs for these models are illustrated in Figures 2,3,4,5,6,7,8, and 9. Table 9 clearly indicates that cubic regression equations have a greater correlation compared to linear regression equations, resulting in more flexible models. But the predicting reliability of cubic equation models may not be the

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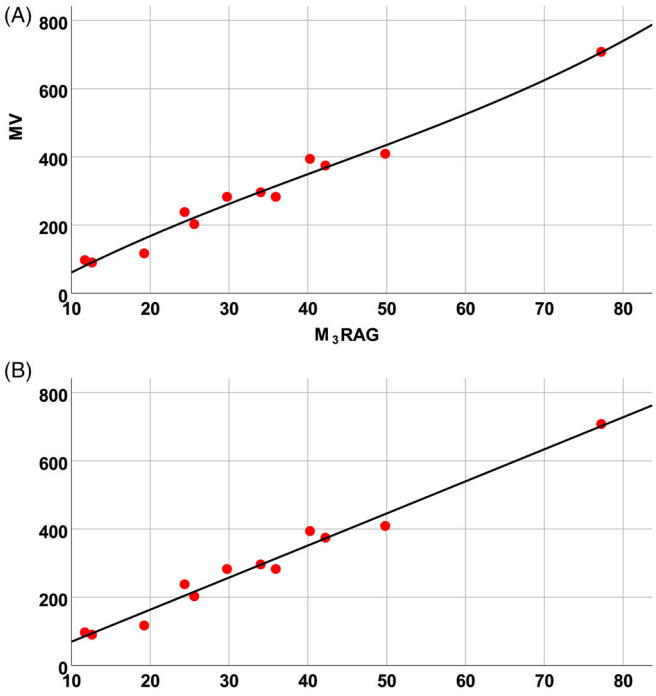


FIGURE 3 Cubic regression graph versus linear regression graph for molar volume. (A) Cubic model. (B) Linear model.

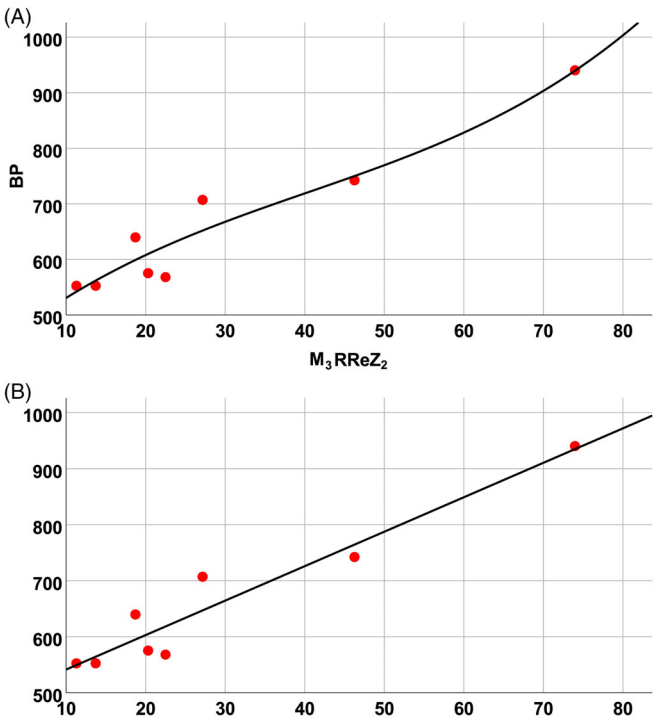


FIGURE 4 Cubic regression graph versus linear regression graph for boiling point. (A) Cubic model. (B) Linear model.

same when extended to larger interval data sets, as evident from Table 9, which shows that cubic regression equations have higher standard errors compared to linear regression equations, and in that case, linear equation models would be extremely helpful. Moreover, as we can see from Figures 2–9, the cubic and linear regression models behave almost the same except for boiling point and enthalpy of vaporization.

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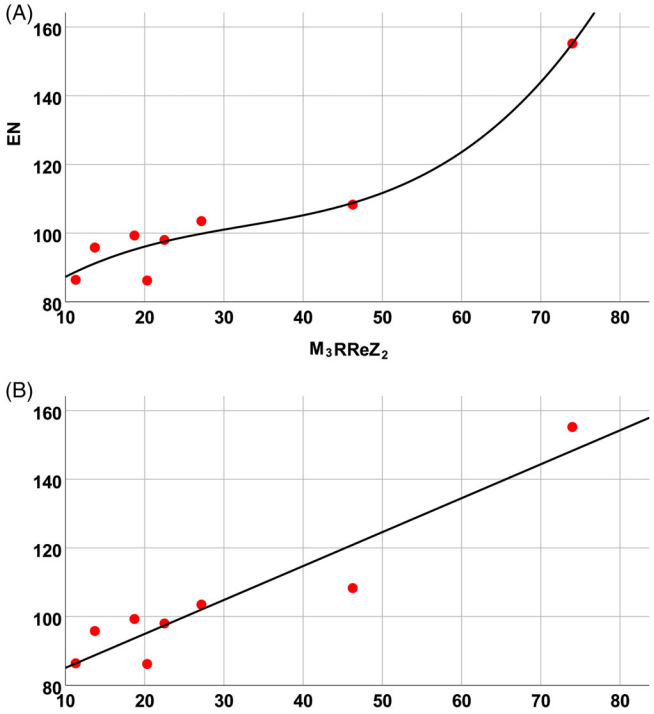


FIGURE 5 Cubic regression graph versus linear regression graph for enthalpy of vaporization. (A) Cubic model. (B) Linear model.

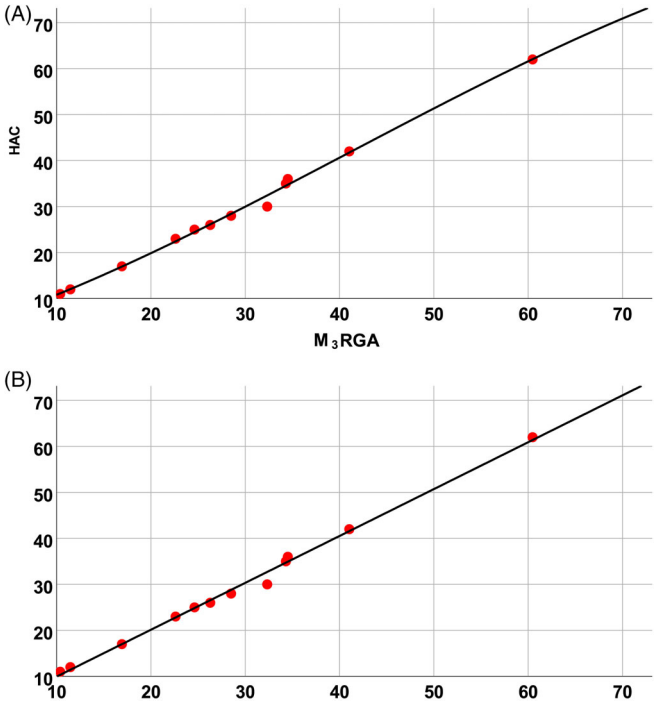


FIGURE 6 Cubic regression graph versus linear regression graph of heavy atom count. (A) Cubic model. (B) Linear model.

We now conclude this section by a comparative analysis of our proposed regression models based on the modified reverse degree over the existing models pertaining to degree and reverse degree [30,31,33,37,45]. Moreover, our proposed models for the data set consist of 12 COVID-19 drug compounds, whereas the existing models mostly focus on eight compounds [30,31,37,45] and the recent one has ten compounds [33]. In general, our models have greater predicting potential for properties such as polarization, molar volume, heavy atom count, molecular refraction,

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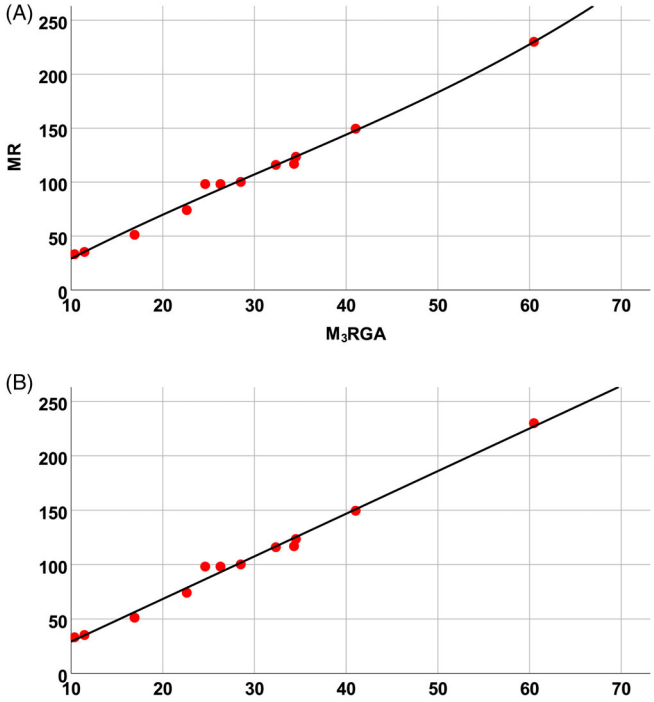


FIGURE 7 Cubic regression graph versus linear regression graph for molar refraction. (A) Cubic model. (B) Linear model.

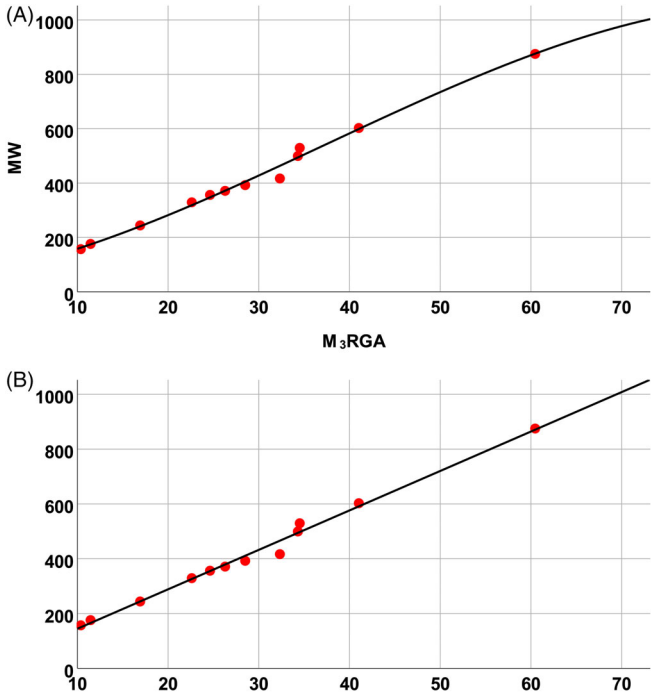


FIGURE 8 Cubic regression graph versus linear regression graph for molar weight. (A) Cubic model. (B) Linear model.

molar weight, and complexity with correlations close to 0.99. Whereas the other properties, boiling point and enthalpy of vaporization have correlations of 0.961 and 0.943, respectively, the models [30] based on the hyper-Zagreb index could be used to predict the properties of boiling point and enthalpy of vaporization.

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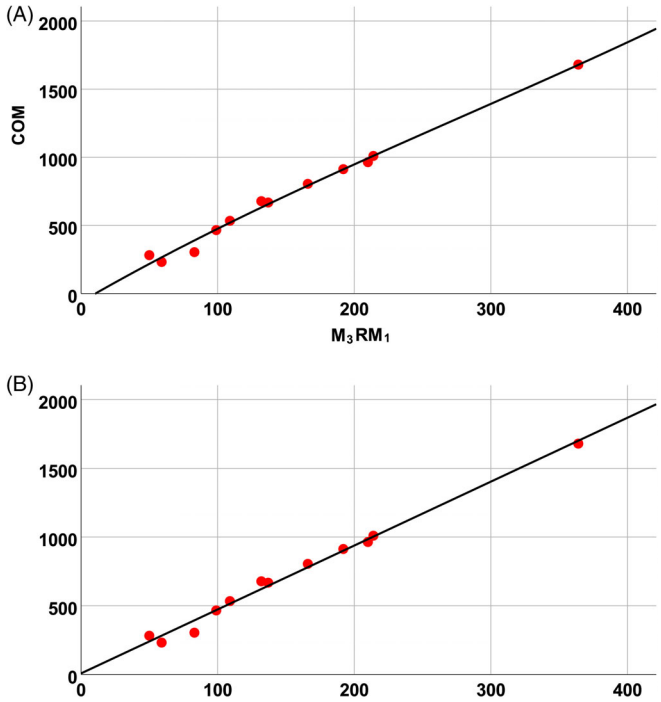


FIGURE 9 Cubic regression graph versus linear regression graph for complexity. (A) Cubic model. (B) Linear model.

TABLE 9 Statistically predicted best regression models with reverse degree parameter $k = 3$.

Properties		Regression equations	r^2	SE
Polarization	Cubic	$POL = 0.000(M_3\Re GA)^3 - 0.018(M_3\Re GA)^2 + 2.022(M_3\Re GA) - 7.034$	**0.992**	2.271
	Linear	$POL = 1.558(M_3\Re GA) - 4.063$	0.991	2.155
Molar volume	Cubic	$MV = 0.001(M_3\Re AG)^3 - 0.120(M_3\Re AG)^2 + 13.636(M_3\Re AG) - 64.472$	**0.978**	30.094
	Linear	$MV = 9.406(M_3\Re AG) - 24.539$	0.977	27.473
Boiling point	Cubic	$BP = 0.001(M_3\Re ReZ_2)^3 - 0.170(M_3\Re ReZ_2)^2 + 11.828(M_3\Re ReZ_2) + 428.048$	**0.927**	48.167
	Linear	$BP = 6.150(M_3\Re ReZ_2) + 479.995$	0.924	40.144
Enthalpy of vaporization	Cubic	$EV = 0.001(M_3\Re ReZ_2)^3 - 0.050(M_3\Re ReZ_2)^2 + 2.027(M_3\Re ReZ_2) + 71.492$	**0.953**	6.300
	Linear	$EV = 0.988(M_3\Re ReZ_2) + 75.197$	0.890	7.879
Heavy atom count	Cubic	$HAC = -8.354E-5(M_3\Re GA)^3 + 0.010(M_3\Re GA)^2 + 0.656(M_3\Re GA) - 3.304$	**0.996**	1.037
	Linear	$HAC = 1.019(M_3\Re GA) + 0.262$	0.995	1.042
Molecular refraction	Cubic	$MR = 0.000(M_3\Re GA)^3 - 0.045(M_3\Re GA)^2 + 5.100(M_3\Re GA) - 17.847$	**0.992**	5.736
	Linear	$MR = 3.919(M_3\Re GA) - 9.977$	0.991	5.392
Molar weight	Cubic	$MW = -0.002(M_3\Re GA)^3 + 0.237(M_3\Re GA)^2 + 6.754(M_3\Re GA) + 69.190$	**0.992**	21.344
	Linear	$MW = 14.391(M_3\Re GA) + 0.664$	0.990	20.440
Complexity	Cubic	$COM = 5.919E-6(M_3\Re M_1)^3 - 0.005(M_3\Re M_1)^2 + 5.791(M_3\Re M_1) - 60.100$	**0.991**	44.897
	Linear	$COM = 4.652(M_3\Re M_1) + 7.656$	0.990	42.516

4 | CONCLUSION

In this paper, we have evaluated the modified reverse degree based topological indices of recently used drugs for COVID-19 by employing the reverse degree edge partition method. We carried out a QSPR analysis on the efficacy of these indices and found a high correlation when the reverse degree parameters increased in nature. From the analysis, we have developed statistically sound models by comparing linear regression and cubic regression to predict the physicochemical properties. It is found that the geometric-arithmetic index is suitable for polarization, heavy atom count, molar refraction, and molar weight properties. The molar volume is best predicted by the arithmetic-geometric index, and the other

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properties such as boiling point, and enthalpy vaporization are best predicted by the redefined Zagreb-2 index. First Zagreb index best predicts complexity. The QSPR analysis of the paper could be helpful in the drug's development against coronaviruses.

AUTHOR CONTRIBUTIONS

Micheal Arockiaraj: Conceptualization; methodology; formal analysis; writing-review and editing. **A. Berin Greeni:** Conceptualization; methodology; supervision; visualization. **AR. Abul Kalaam:**Methodology; investigation; validation; writing-original draft.

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CONFLICT OF INTEREST STATEMENT

The authors declare that they have no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data used to support the findings of this study are included in the manuscript. Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

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